

Band-limited Vorticity Direction: Corrected Estimates and a Sharpness Obstruction

Working correction for Jonathan Robert Simons
Prime Field Technologies LLC

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Scope notice (delivery wrapper): spatial geometry only; no PDE / Navier–Stokes / Clay regularity claim.

Abstract

This note replaces the geometric estimate in the recovered June 19 Ring Lemma manuscript. For a three-dimensional band-limited vorticity field, a threshold relative to its L^2 norm gives a direction-gradient bound of order frequency to the power $5/2$. An explicit divergence-free family shows that this power cannot be reduced at a fixed threshold. A threshold relative to the L^∞ norm instead gives a bound linear in frequency. These are spatial estimates; no dynamical preservation or Navier–Stokes regularity theorem is asserted. Novelty has not been audited.

Work on $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ with normalized measure $dx/(2\pi)^3$. All norms below use this measure. Let u be a real, mean-zero, divergence-free trigonometric polynomial, $\omega = \nabla \times u \neq 0$, and $\xi = \omega/|\omega|$ where $\omega \neq 0$. Suppose its Fourier support lies in $|k| \leq L$, $L \geq 1$.

Proposition 1 (L^2 -threshold estimate). *For $0 < c < 1$, set $E_c = \{x : |\omega(x)| \geq c\|\omega\|_2\}$. Then $|E_c| > 0$, and*

$$\|\nabla \xi\|_{L^\infty(E_c)} \leq \frac{C}{c} L^{5/2}.$$

The constant depends only on the torus normalization and the frequency cutoff convention.

Proof. At nonzero vorticity,

$$\partial_\ell \xi = \frac{(I - \xi \otimes \xi) \partial_\ell \omega}{|\omega|}, \quad |\nabla \xi| \leq \frac{|\nabla \omega|}{|\omega|}.$$

Fourier Cauchy–Schwarz and Parseval give

$$\|\nabla \omega\|_\infty \leq \left(\sum_{|k| \leq L} |k|^2 \right)^{1/2} \|\omega\|_2 \leq CL^{5/2} \|\omega\|_2.$$

Dividing on E_c proves the derivative bound. Also,

$$\|\omega\|_2^2 \leq c^2 \|\omega\|_2^2 + \|\omega\|_\infty^2 |E_c|,$$

so $|E_c| \geq (1 - c^2) \|\omega\|_2^2 / \|\omega\|_\infty^2 > 0$. This lower bound need not be uniform in L . □

Proposition 2 (Fixed-threshold sharpness). *There are real, smooth, divergence-free velocity fields supported in a single dyadic annulus for which*

$$\|\nabla \xi\|_{L^\infty(E_{1/2})} \geq aL^{5/2}$$

with $a > 0$ independent of L . Consequently a uniform CL bound on the original L^2 -threshold set is false.

Proof. Let $n \geq 4$ be a power of two and define

$$K_n = \{(k_1, k_2, k_3) : n \leq k_1 \leq 2n - 1, 0 \leq k_2, k_3 \leq n/4\}, \quad M = |K_n| = n(n/4 + 1)^2.$$

With $P_k = I - k \otimes k/|k|^2$ and the coordinate vectors e_2, e_3 , put

$$\omega_n(x) = e_3 \cos(nx_1) + M^{-1/2} \sum_{k \in K_n} P_k e_2 \sin(k \cdot x).$$

This field is divergence-free and mean-zero. It is a curl: define $\hat{u}_n(k) = ik \times \hat{\omega}_n(k)/|k|^2$ for $k \neq 0$. All frequencies satisfy $n \leq |k| < 4n$, a fixed dyadic annulus; take $L = 4n$. Orthogonality of the sine modes and the cosine gives

$$\|\omega_n\|_2^2 = \frac{1}{2} + \frac{1}{2M} \sum_{k \in K_n} |P_k e_2|^2 \leq 1.$$

At the origin, $\omega_n(0) = e_3$, so the origin lies strictly inside $E_{1/2}$. Moreover

$$e_2 \cdot \partial_1 \xi_n(0) = M^{-1/2} \sum_{k \in K_n} k_1 \left(1 - \frac{k_2^2}{|k|^2}\right) \geq \frac{15}{16} n \sqrt{M} \geq \frac{15}{64} n^{5/2}.$$

Continuity on a neighborhood of the origin makes this a lower bound for the essential supremum on $E_{1/2}$. Replacing n by $L/4$ gives the stated absolute constant. $\square$

Proposition 3 (Linear estimate under an additional amplitude condition). *For $0 < a \leq 1$, define $F_a = \{x : |\omega(x)| \geq a\|\omega\|_\infty\}$. Then*

$$\|\nabla \xi\|_{L^\infty(F_a)} \leq \frac{C}{a} L.$$

Alternatively, if $\|\omega\|_\infty \leq A\|\omega\|_2$, then

$$\|\nabla \xi\|_{L^\infty(E_c)} \leq \frac{CA}{c} L.$$

Proof. The band-limited Bernstein estimate $\|\nabla \omega\|_\infty \leq CL\|\omega\|_\infty$, combined with the direction derivative identity, gives both conclusions. It follows, for example, by convolving with a smooth Fourier cutoff equal to one on $|k| \leq L$, whose differentiated kernel has L^1 norm at most CL . $\square$

Remark 1 (Scope). *The additional ratio A is a spatial amplitude-control hypothesis. Frequency localization alone gives only $A \leq CL^{3/2}$. Neither this hypothesis nor a bound on F_a is shown here to propagate under Navier–Stokes evolution. Single-shell support is generally not invariant under nonlinear convolution. Bounds on a moving high-vorticity subset must be matched to the exact hypotheses of a geometric regularity criterion before drawing a PDE conclusion.*